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BACHARELADO EM

ENGENHARIA DE SOFTWARE

ACCELERATION OF PATTERN MATCHING ALGORITHMS USING PARALLEL PROGRAMMING

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Brasília - DF, 2025

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Trabalho de Conclusão de Curso apresentado como requisito parcial para a obtenção de grau de Bacharel em Engenharia de Software, pelo Instituto Brasileiro de Ensino, Desenvolvimento e Pesquisa (IDP).

Orientador
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ABSTRACT

O Abstract é a tradução do resumo para a língua inglesa. Ele tem exatamente a mesma função e estrutura: apresentar, de forma breve e precisa, os elementos essenciais do trabalho — tema, objetivo, metodologia, resultados e conclusões. Assim como no resumo, o abstract deve ser elaborado em um único parágrafo, sem citações, com linguagem formal, objetiva e impessoal. Ao final do abstract, também devem ser inseridas as keywords, que são as palavras-chave em inglês.

Keywords: LaTeX, metodologia científica, trabalho de conclusão de curso, paper, technology.

RESUMO

O Resumo é uma apresentação concisa do conteúdo do trabalho. Deve sintetizar, de forma clara e objetiva, os principais elementos do texto: tema, objetivo, metodologia, resultados e conclusões. O objetivo do resumo é permitir que o leitor compreenda rapidamente o que foi desenvolvido no trabalho, sem precisar lê-lo na íntegra. De acordo com as normas da ABNT (NBR 6028), o resumo deve ser elaborado em um único parágrafo, sem citações, com a utilização de linguagem impessoal e verbo na voz ativa. Geralmente, possui entre 150 e 500 palavras, dependendo do regulamento da instituição. Além do texto, o resumo deve apresentar de três a cinco palavras-chave, separadas por ponto e finalizadas com ponto final, que representem os principais assuntos abordados no trabalho.

Palavras-chave: LaTeX, metodologia científica, trabalho de conclusão de curso, artigo, tecnologia.

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1

1 INTRODUCTION

Advancements in semiconductor technology have historically driven computational power growth. For decades, the performance of single-core processors followed the trend predicted by *Moore's Law*, which forecast the doubling of transistors on a chip every eighteen months [1]. However, in the mid-2000s, the hardware industry encountered significant physical barriers, such as excessive power consumption and challenges in heat dissipation, which limited the increase in processor clock frequency. The response to these technological limitations was a change in processor design: instead of focusing on making a single core faster, the emphasis shifted to integrating multiple processing cores onto a single chip. Thus, multi-core processors emerged as the dominant architecture in modern computers. This transition opened up new possibilities for performance enhancement; however, to take advantage of the performance gains offered by multi-core processors, programs must be intentionally developed or adapted to execute tasks in parallel.

1.1 CONTEXT AND PROBLEM STATEMENT

This architectural shift from sequential to parallel execution profoundly impacts high-performance applications. In information security, tools for malware analysis and network intrusion detection must process vast data volumes rapidly to identify malicious patterns and enable prompt responses. YARA, an industry standard for pattern-based malware classification, exemplifies this challenge. It utilizes the efficient Aho-Corasick algorithm as the core of its multi-pattern matching engine, allowing it to search thousands of string patterns simultaneously [2].

Despite its efficiency for multi-pattern searching, YARA's internal implementation of the Aho-Corasick algorithm is predominantly sequential. While YARA supports *inter-file parallelism*—concurrently scanning multiple smaller files across different threads—it lacks *intra-file parallelism*. This means that when scanning a single, large target (e.g., a full disk image or a large network packet capture), the entire computational load falls upon a single CPU core. This limitation constitutes a significant performance bottleneck in critical, real-world security scenarios, where rapid analysis of large-volume targets is paramount. As the size of data in security incidents continues to grow, this bottleneck becomes increasingly prohibitive.

1.2 JUSTIFICATION AND RESEARCH PROPOSAL

The identified performance limitation of YARA's sequential engine when scanning large files presents a clear opportunity for optimization through parallel computing. Scientific literature confirms that parallelizing the Aho-Corasick algorithm is a viable and effective strategy; previous studies in fields like bioinformatics report significant speedups [3]. However, a systematic review of the literature reveals that direct implementation, integration, and empirical evaluation of a parallel Aho-Corasick engine within the YARA framework remains an unexplored research area.

Therefore, this research proposes a *data-parallel scanning strategy* for YARA's Aho-Corasick engine on multi-core CPUs. The proposed solution involves partitioning each large input file into fixed-size, overlapping chunks and using multiple `POSIX threads` to concurrently traverse the same pre-compiled Aho-Corasick automaton. Each thread will process its assigned chunk, and thread-local match lists will be merged upon scan completion. The originality of this work lies not in inventing a new parallel algorithm, but in the software engineering required to integrate this strategy into YARA's complex `C`-based codebase and the scientific rigor of its empirical evaluation.

1.3 OBJECTIVES

The general objective of this thesis is to demonstrate that the performance of YARA's pattern matching engine can be significantly accelerated for large-file scanning scenarios by introducing *intra-file parallelism* on multi-core CPU architectures.

To achieve this general objective, the following specific objectives are defined:

1. To design and implement a parallel version of the Aho-Corasick scanning loop within the `libyara` library using native threading techniques (`POSIX threads`).

2. To solve the technical challenges associated with this integration, including thread-safe result management, synchronization overhead, and the handling of pattern matches that cross chunk boundaries.

3. To conduct a rigorous experimental evaluation to quantify the performance gains of the parallel implementation in terms of speedup and throughput.

4. To analyze the scalability of the proposed solution by systematically varying key parameters, such as the number of threads, input file size, and the complexity of the YARA rule set.

1.4 STRUCTURE OF THE THESIS

This thesis is organized into the following chapters. Chapter 2 presents the Theoretical Framework, covering the fundamentals of parallel computing, including architectures, metrics, and models for parallel programming with threads. It also details the core data structures and algorithms relevant to this work, specifically Tries, the general problem of pattern matching, and a comprehensive examination of the Aho-Corasick algorithm. Chapter 3 outlines the Systematic Literature Review process, detailing the research questions, search strategy, and selection criteria used to identify the state of the art and establish the research gap. Chapter 4 describes the Methodology…

2

2 THEORETICAL FRAMEWORK

2.1 FUNDAMENTALS OF PARALLELISM

Parallelism in computing refers to the simultaneous execution of multiple tasks or parts of the same task, with the primary goal of reducing a program's total execution time. It is crucial to distinguish parallelism from concurrency. Concurrency occurs when multiple tasks are in progress at a given time, but may not be executing simultaneously. A single-core system can exhibit concurrency by rapidly switching between the execution of different tasks, simulating parallelism through preemption. True parallelism, on the other hand, requires hardware with multiple processing units [4].

2.1.1 Parallel Computing Architectures

Parallel computer architectures are classified according to various criteria. A prominent taxonomy, proposed by Flynn, categorizes these architectures based on instruction and data streams [5]:

- **SISD (Single Instruction, Single Data)**: Represents a conventional single-processor architecture, where a single instruction operates on a single data stream.
- **SIMD (Single Instruction, Multiple Data)**: Involves multiple processing units that execute the same instruction concurrently on different data elements.
- **MISD (Multiple Instruction, Single Data)**: Characterizes systems where multiple processing units apply distinct instructions on the same data stream.
- **MIMD (Multiple Instruction, Multiple Data)**: Involves multiple processing units that execute distinct instructions on distinct data streams independently. This paradigm is directly related to multi-core architectures.

Within the MIMD category, systems are further distinguished based on their memory organization [1]:

- **Shared-Memory Systems**: In these systems, multiple processors share a physical memory address space. Inter-processor communication (IPC) is typically achieved by reading from and writing to shared variables in memory or through

message passing mechanisms. Multi-core processors are the most common and widely available example of this architecture, where all cores on a chip share access to main memory.

- **Distributed-Memory Systems**: Each processor in these systems possesses their own locally accessible memory, and communication between processors is usually done by exchanging messages across an interconnection network. Computer clusters are examples of this architectural model.

The present work focuses on the acceleration of algorithms in environments with multi-core processors, which fall into the shared memory MIMD category. A multi-core processor (MCP) is a single chip that contains two or more central processing units (CPUs) called 'cores'. Each of these cores is capable of reading and executing program instructions independently, essentially functioning as a complete processor.

2.1.2 Parallel Programming With Threads

Within shared-memory systems, such as multi-core processors, software-level parallelism is commonly achieved through the execution of multiple threads within a single process.

Processes vs. Threads

A process is an instance of an executing program, an entity possessing its own dedicated virtual address space, file descriptors, environment variables, and other Operating System (OS) resources, all isolated from other processes [1].

A thread, on the other hand, is a lighter unit of execution started by a process. The same process can control, concurrently or in parallel, multiple threads. A key characteristic is that all threads within a process share its virtual address space, which encompasses executable code, global data, and the heap segment. However, each thread maintains its private resources, which are essential for its independent execution context:

- A unique identifier (thread ID).
- A set of CPU registers, including the program counter (PC).
- A separate execution stack, used for local variables and function call control.
- Thread-specific state, such as signal masking and scheduling priority.

POSIX Thread Model (Pthreads)

The POSIX Thread Model, commonly referred to as "Pthreads", is a standard that defines an application programming interface (API) for thread creation and management, as specified by IEEE Std 1003.1c-1995. Pthreads provides a portable suite of data types, functions, and constants that allows developers to create multithreaded

applications that are compilable and executable across diverse POSIX-compliant operating systems (e.g., Linux, macOS, and various Unix variants). The Pthreads API provides functionality for managing the thread lifecycle—including creation, termination, and joining—and, critically, incorporates synchronization primitives. These primitives, such as mutexes and condition variables, are indispensable for orchestrating access to shared resources within shared-memory environments [1, 4].

2.1.3 Parallel Performance Evaluation Metrics

Performance evaluation is fundamental to understanding the effectiveness of a parallel program. The following metrics compare the execution time of the parallel program with its equivalent sequential counterpart [6].

Execution Time and Overhead

The performance of a program is measured by its execution time. For parallel programs, the following terms are defined:

- **Ts**: The execution time of a baseline sequential algorithm executed on a single processor unit.
- **Tp**: The execution time of the parallel program utilizing p processors.

Ideally, `Tp` is expected to be substantially less than `Ts`. However, parallel execution introduces *overhead*. This encompasses the time spent by processors on tasks that do not directly contribute to problem resolution, including communication, synchronization, load balance, and other computations inherent to the parallel implementation.

Speedup

Speedup (`S`) measures the performance gain achieved through parallel execution relative to sequential execution and it is referenced as Equation 1. It is defined as the ratio of sequential execution time to parallel execution time:

$$S = \frac{T_s}{T_p} \tag{1}$$

An ideal, or linear, speedup would be `S = p`. Amdahl's Law, establishes a theoretical upper bound on the speedup achievable by parallelizing a program with a fixed problem size. `s` represents the fraction of a program's sequential execution time that is inherently serial (non-parallelizable), and `f` is the parallelizable fraction (`s + f = 1`), the maximum speedup on `p` processors is given by:

$$S \leq \frac{1}{s + f/p} \tag{2}$$

In the limit, as the number of processors (`p`) tends to infinity, the maximum speedup is bounded by `1/s`. This implies that even a small sequential fraction (`s` large) can

severely limit the total speedup, regardless of the number of processors available. A complementary perspective is offered by the Gustafson-Barsis Law. This law argues that a speedup approximately linear

$$S_{\text{escalado}} = s + (1 - s) \cdot p \tag{3}$$

is achievable for problems that benefit from this scalability. This formulation suggests that for problems that can be scaled, it is possible to obtain speedup approximately linear with the number of processors, even in the presence of a serial fraction. Both Amdahl's and Gustafson's laws are important for understanding the limits and potential of parallelism in different scenarios.

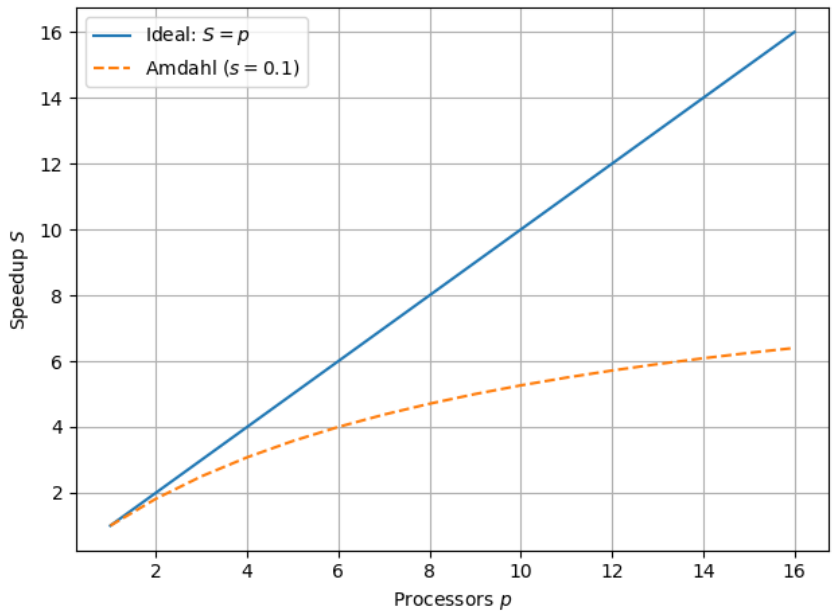

Figure 1: Ideal vs. Amdahl's theoretical speedup ($s = 0.1$).

Efficiency

Efficiency (E) measures the effectiveness of processor utilization in a parallel program. It is defined as the ratio of speedup to the number of processors p:

$$E = \frac{S}{p} = \frac{T_s}{p \cdot T_p} \tag{4}$$

Efficiency values range between 0 and 1 (inclusive), often expressed as a percentage (0% to 100%). An efficiency of 1 (or 100%) signifies ideal, linear speedup, implying that all processors are contributing productively throughout the execution. In practice, efficiency typically diminishes as the processor count increases, primarily due to the escalating impact of parallel overhead. **Scalability**

Scalability, in the context of a parallel program and its underlying architecture, de-

notes its capacity to sustain performance (often measured by efficiency) as the number of processors and/or the problem size increases.

Strong Scalability: Measures how execution time varies with an increasing number of processors for a fixed total problem size. The objective is to maintain relatively constant efficiency as $\mathtt{p}$ increases. Amdahl's Law is particularly relevant to this scenario.

Weak Scalability: Evaluates performance as both the number of processors $\mathtt{p}$ and the total problem size $\mathtt{W}$ increase, such that the problem size per processor $\mathtt{W}/\mathtt{p}$ remains constant. The goal is to maintain efficiency as $\mathtt{p}$ and $\mathtt{W}$ scale proportionally. Gustafson's Law is most pertinent in this context.

A system is considered scalable if the efficiency remains above a certain reasonable limit as the platform (number of processors) and the problem grow.

2.2 TRIES (PREFIX TREES)

This section elucidates the Trie data structure, detailing its defining properties, core operational mechanics, performance aspects, and its significance as a foundational element in advanced string processing algorithms, particularly the Aho-Corasick algorithm.

2.2.1 Definition and Basic Structure

A Trie—also known as a prefix tree—is a specialized tree-like data structure optimized for the efficient storage and retrieval of a dynamic set of strings. Edward Fredkin introduced the term "trie" in 1960, highlighting its primary application in information retrieval [7, 8]. Figure 2 illustrates an example of a trie storing a set of words.

Tries store strings by organizing nodes so that each path from the root to a node represents a unique prefix. In a standard R-way trie, each node can branch to R children, where R denotes the size of the character alphabet. The existence of a parent-child edge highlights two key aspects of tries:

- The edge's position (index) within the parent node's array of R possible child links directly corresponds to a unique character in the alphabet. For instance, index 0 might correspond to 'a', index 1 to 'b', and so on, according to a predefined mapping.
- The character associated with this edge extends the prefix represented by the parent; this implies that at least one word (or key) in the trie contains this character sequence.

To signify that such a prefix also constitutes a complete key present in the trie, the node reached at the end of this path is typically marked by storing an associated non-null value or a boolean flag. The characters themselves are implicitly encoded by the trie's structure rather than being explicitly stored within each node.

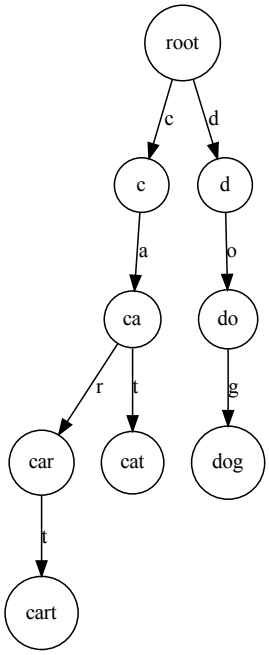

Figure 2: Example trie storing {“car”, “cat”, “cart”, “dog”}. Nodes marking the end of a complete word are shown with a double circle.

2.2.2 Core Operations

Tries support several primary operations, including search, insertion, and deletion [9].

Search: Locating a key involves traversing the trie starting from the root, following the successive characters of the target key. Each character dictates which child link to follow. The search operation succeeds if a complete path corresponding to all characters of the key exists and the node reached at the end of this path is appropriately marked as representing a complete key. The search fails if, at any point, a required link is null (i.e., no path for the current character exists) or if the path is fully traversed but the final node is not marked as a complete key.

Insertion: The insertion process begins similarly to a search, by traversing the trie according to the characters of the key to be added. If the path for the key does not extend to its full length (i.e., a null link is encountered), new nodes are created and linked to form the remainder of the path. The node corresponding to the final character of the inserted key is then marked to indicate that it is a complete key in the set.

Deletion: Deleting a key first involves locating the node corresponding to the key’s final character and removing its end-of-word marker. If there are no child links (meaning it is not a prefix for any other key), it can be deleted. This deletion may proceed recursively: when a child node is removed, if its parent has no other children and the parent

node is not marking a key, the parent node becomes redundant and can be deleted. This process continues up to the root as long as parent nodes become redundant.

2.2.3 Advantages and Performance Characteristics

Tries provide distinct advantages for operations on string collections.

Time Complexity: A primary advantage is that the time required for search and insertion operations depends primarily on the length of the key (L), not on the total number (N) of keys stored. Specifically, these operations typically require examining a number of nodes proportional to L. Sedgewick & Wayne (2011) state that the number of array accesses (node visits) is at most $L + 1$ [9]. This $O(L)$ performance makes tries highly efficient for prefix-based searches (e.g., retrieving all keys starting with a given prefix) and ensures that search performance does not substantially degrade as the dataset size increases.

Prefix Sharing: When stored strings share common prefixes, tries inherently exploit this by having those strings share the initial path from the root. This sharing can lead to significant space savings compared to storing each string independently, particularly if there is substantial prefix overlap among keys [7].

2.2.4 Space Complexity Considerations

While prefix sharing can save space, the overall space complexity of R-way tries requires careful consideration. Each node in an R-way trie conceptually provides R potential links. If the alphabet size R is large (e.g., for Unicode) and the actual branching factor at most nodes is low (a sparse trie), allocating space for all R links per node can lead to considerable memory consumption due to numerous unused (null) links [8, 9].

2.2.5 Foundational Role in String Processing

The trie data structure is a cornerstone for numerous advanced string processing algorithms. Critically for this work, the Aho-Corasick algorithm, engineered for the efficient, simultaneous matching of multiple patterns within a text, employs a trie—constructed from the set of patterns (the dictionary)—as its initial structural framework. This pattern trie is subsequently augmented with specialized ”failure links” to form a finite automaton, enabling text processing in time linear to the text's length [10]. A thorough grasp of trie principles is thus indispensable for developing and analyzing the Aho-Corasick algorithm.

2.3 PATTERN MATCHING

This section introduces the fundamental problem of pattern matching within strings, explores key algorithmic approaches to its solution, and underscores its significance across various computational domains. This foundational discussion provides the essential context for understanding advanced multi-pattern matching algorithms, such as

the Aho-Corasick algorithm.

2.3.1 The Pattern Matching Problem

Pattern matching involves identifying all instances where a given pattern P, which is a string of length M, appears as a contiguous substring within a larger text T, which has a length of N. Beyond simply finding the first match, the problem often extends to locating all occurrences, enumerating them, or extracting surrounding contextual information for each match [9].

2.3.2 Importance and Applications

Pattern matching stands as a cornerstone problem in computer science with diverse applications. In general information processing, it underpins the functionality of text editors and web search engines for keyword searches. The field of computational biology extensively utilizes pattern matching for analyzing genetic sequences [9].

Furthermore, in the field of information security, pattern matching is crucial: Network Intrusion Detection Systems (NIDS) utilize it to identify malicious signatures in network traffic. Tools like YARA, for instance, define rules based on textual or binary patterns to classify and identify malware [2]. Later in this work, we'll explore the role of pattern matching for NIDS in greater detail.

2.3.3 Pattern Matching Algorithms

Pattern matching algorithms are generally categorized based on the number of patterns they process and the exactness of the match. This research focuses on Exact Matching, where the pattern must correspond precisely to a segment of the text. A key distinction for this study lies between algorithms designed for Single-Pattern Matching and those optimized for Multi-Pattern Matching, where the objective is to identify all occurrences of any pattern from a predetermined set of patterns within a given text [10, 11].

2.3.4 Fundamental Algorithms for Exact Single-Pattern Matching

The Naive Algorithm

The naive or brute-force algorithm represents the most straightforward method for exact pattern matching. Its operation involves iteratively aligning the pattern P with the text T at every possible starting position i (from 0 to $N - M$). For each alignment, P is compared character by character, from left to right, against the corresponding substring. If all characters match, an occurrence is reported [11].

Following this comparison (whether a match or a mismatch), the algorithm shifts P one position to the right relative to T and proceeds to the next alignment. Despite its simplicity, Knuth, Morris, and Pratt (1977) state that this method can be highly inefficient [12]. Its worst-case time complexity is $O(M \cdot N)$ character comparisons. This occurs, for example, when searching for a pattern like $P = a^n b$ within a text $T = a^{2n} b$, where many partial matches occur.

Optimized Approaches via Pattern Preprocessing

To address the limitations of the naive algorithm, more advanced pattern matching algorithms include a preprocessing phase for the pattern P. This phase extracts structural information about P that is then utilized during the search, enabling larger shifts of the pattern relative to the text and thus reducing the total number of character comparisons.

Knuth-Morris-Pratt (KMP) Algorithm: The KMP algorithm [12] efficiently resolves the pattern matching problem with an optimal worst-case time complexity of $O(N+M)$. Unlike the naive approach, KMP avoids re-checking characters that have already been matched. To achieve this, KMP first preprocesses the pattern P by constructing a special array known as the failure array (or prefix array), typically denoted by $p[1..M]$. This array indicates the length of the longest prefix that is also a suffix for each prefix of P. With this information, when a mismatch occurs during the search, the algorithm utilizes the failure array to determine how far the pattern should shift to the right, bypassing unnecessary comparisons and preventing redundant checks.

Boyer-Moore (BM) Algorithm: The Boyer-Moore (BM) algorithm [13] is widely recognized for its excellent practical performance, often outperforming linear-time methods in typical cases. Unlike the KMP algorithm, Boyer-Moore typically compares characters starting from the end of the pattern and moves backward. Its efficiency primarily stems from two rules (or heuristics):

- **Bad Character Rule**: When a mismatch occurs, this rule uses the character from the text that caused the mismatch to determine how far the pattern can safely shift. The algorithm shifts the pattern to align this mismatched text character with its last occurrence in the pattern or moves completely past the mismatched character if it does not appear in the pattern.

- **Good Suffix Rule**: When a mismatch happens after matching one or more characters from the end of the pattern, this rule uses the matched suffix to decide how far to shift the pattern. It searches for another occurrence of the matched suffix within the pattern, ensuring that no previously matched characters are unnecessarily compared again.

These two rules combined enable the Boyer-Moore algorithm to skip larger portions of the text, significantly reducing the total number of comparisons.

2.3.5 Limitations of Iterating Single-Pattern Algorithms

When the task involves searching for multiple patterns simultaneously from a set $K = \{P_1, P_2, \ldots, P_k\}$ within a text T, simply applying a single-pattern matching algorithm for each P_i independently proves highly inefficient. This ”iterative” approach requires scanning the text multiple times, resulting in a total time complexity that can be

approximately $O(k \cdot (N + M_{\text{avg}}))$ or worse, where M_{avg} is the average pattern length [10]. This highlights the need for more specialized solutions.

2.3.6 The Need for Specialized Multi-Pattern Algorithms

To efficiently handle applications that require simultaneous searches for a large number of patterns, specialized algorithms are essential. These methods usually preprocess the entire set of patterns to create a unified data structure, which enables a single, efficient pass over the text to identify all occurrences of any pattern within the set.

The theoretical framework of *finite automata* provides a robust and elegant solution for this problem. By constructing a single *finite automaton* (FA) that can recognize all patterns in a given set, the text can be processed in a single continuous scan. Each character read from the text triggers a *state transition* within the FA. When the FA reaches an *accepting state*—a state explicitly designed to indicate the successful recognition of one or more patterns—the algorithm reports an occurrence. Algorithms like the Aho-Corasick algorithm exemplify this approach; they automate the construction of such a *pattern-matching machine*, typically based on a trie structure augmented with specialized *failure transitions* [10]. This automaton-based paradigm forms the critical foundation for efficiently solving multi-pattern matching problems.

2.4 THE AHO-CORASICK ALGORITHM

This section provides a comprehensive examination of the Aho-Corasick algorithm, a highly efficient string-matching method designed to locate all occurrences of multiple patterns (keywords) within a given text. It combines *automata theory* with *trie* data structures to efficiently solve the *multi-pattern search* problem. Its search phase runs in $O(n + z)$ time, where n is the text length and z is the number of matches [10].

2.4.1 Context

The task of concurrently identifying all instances of a large collection of patterns within a text poses a significant computational challenge. A naïve approach—running a *single-pattern matching* algorithm such as KMP or BM for each of the k patterns—requires repeated scans of the text and has a worst-case time complexity of $O(k \cdot n)$, where k is the number of patterns and n is the text length, rendering it impractical for large pattern sets or long texts [11].

The Aho-Corasick algorithm overcomes this by building one *finite automaton* that encodes all patterns simultaneously. This automaton processes the input text in a single pass to detect all matches. The algorithm effectively combines the *state-transition* efficiency of finite automata with the prefix-sharing advantages of tries, following *failure links* on mismatches and resuming matching without rescanning characters—an extension of KMP's *failure-function* idea to a trie of patterns.

2.4.2 Algorithm Components and Construction

Aho-Corasick has two main phases: (i) construction, which builds the automaton, and (ii) search, which scans the text. Three key tables, built in the following order, define the automaton:

2.4.3 Keyword Trie (goto)

A *keyword trie* is first constructed from the pattern set $K = \{P_1, P_2, \ldots, P_k\}$. Each node represents a state, and each edge corresponds to a character transition. A path from the root to any state s spells a prefix of one or more keywords. Formally, the *goto function* goto(s, a) returns the next state s' when an edge labelled a exists; otherwise the transition fails. Undefined transitions from the root are assigned back to the root, i.e., goto$(0, a) = 0$, creating implicit self-loops for every alphabet symbol.

2.4.4 Failure Function (f)

For any state s (except the root), f(s) is the state s' whose path label is the longest proper suffix of the path label of s that is also a prefix of some keyword. On a mismatch, the automaton follows f(s) and resumes matching without consuming the current character. The *failure function* is computed level-by-level (breadth-first) after the goto table is complete.

2.4.5 The Output Function

The *output function*, output(s), identifies all keywords that end at state s. For a given state s, output(s) is the set of all patterns P_i in K such that the string corresponding to the path to s is P_i. To ensure all matches are reported, including those that are suffixes of other patterns, output(s) is augmented during the failure function computation: if f$(s) = s'$, then output$(s) \leftarrow$ output$(s) \cup$ output(f(s)). This incremental union lets each state report every pattern that ends either at that state or at any suffix state reached through its failure chain.

2.4.6 Matching Phase

Once the Aho-Corasick automaton is constructed, it processes the input text T character by character in a single pass. Starting from the root state, the automaton iterates over each text character a_i as follows:

1. Attempt goto(*current_state*, a_i) for the current character a_i.
2. If the transition succeeds, set *current_state* $\leftarrow s'$.
3. Otherwise, set *current_state* $\leftarrow$ f(*current_state*) and repeat step 1 without consuming a_i until a goto transition succeeds or the root is reached. From the root, goto$(0, a_i)$ always exists, possibly returning to the root itself.

4. After each successful transition, inspect output(*current_state*) and report every pattern it contains, recording its ending position i.

The combined number of goto and failure transitions during a scan of length n is $< 2n$ [10].

2.4.7 Time and Space Complexity

The Aho-Corasick algorithm exhibits optimal *linear time complexity* for multi-pattern matching [10, 11].

- **Construction phase**: building the goto, failure, and output functions takes $O(L)$ time, where L is the total length of all keywords. If a dense array is used for the root, an additional $O(|\Sigma|)$ initialization cost is incurred, giving $O(L + |\Sigma|)$.
- **Search phase**: Scanning a text of length n performs fewer than $2n$ state transitions. Reporting matches costs time proportional to the total number of occurrences z, so the total search time is $O(n + z)$.
- **Space complexity**: Storing the automaton requires $O(L + |\Sigma|)$ space: $O(L)$ for trie edges, $O(L)$ pointers for failure links, and additional space for output lists (proportional to the number of stored outputs).

2.4.8 Applications and Relevance

The Aho-Corasick algorithm is widely used in applications that require high-performance *multi-pattern matching*. Its single-pass processing capability makes it ideal for systems such as Network Intrusion Detection Systems (NIDS), which scan network traffic against thousands of malicious signatures [10]. In cybersecurity, it is a core component of anti-malware tools and engines, such as YARA, where it efficiently matches literal string components extracted from complex rules. Other applications include large-scale text analysis, motif searching in DNA or protein sequences, and spam filtering.

While Aho-Corasick excels at exact literal matching, systems like NIDS often layer additional capabilities—such as *regular-expression matching*—on top of Aho-Corasick engines to support more expressive signatures, which affects overall performance characteristics. The algorithm therefore remains a theoretically elegant and practically impactful example of *finite-automata theory* applied to string processing [2, 11].

3

3 SYSTEMATIC REVIEW

This section outlines the methodology employed for the Systematic Literature Review (SLR) conducted to identify, evaluate, and synthesize pertinent studies concerning the parallelization and optimization of the Aho-Corasick algorithm. The review focuses on implementations targeting multi-core CPUs. The objective of this SLR is to support the development of the present research and to answer the formulated research questions.

3.1 RESEARCH QUESTIONS

The following Research Questions (RQs) were formulated to guide the literature search, selection, and analysis process:

1. **RQ1:** What parallelization techniques and associated challenges are documented in the literature for implementing the Aho-Corasick algorithm utilizing threads on multi-core CPUs?

2. **RQ2:** What specific optimizations for the Aho-Corasick algorithm are employed in application scenarios involving extensive pattern sets, particularly within domains of information security?

3.2 SEARCH STRATEGY AND STUDY SELECTION PROTOCOL

The search and selection protocol for identifying primary studies encompassed the definition of data sources, construction of search queries, and establishment of inclusion and exclusion criteria.

3.2.1 Data Sources

As seguintes bases de dados científicas foram selecionadas devido à sua abrangência e relevância na área de Ciência da Computação e Engenharia de Software:

- Web of Science (WoS)
- Scopus
- ACM Digital Library (ACM DL)

- IEEE Xplore

3.2.2 Search Queries

Two primary search strings were constructed to address the aspects of parallelization (RQ1) and optimization/application (RQ2) pertinent to the Aho-Corasick algorithm. The literature search was executed in May 2025.

- **String 1 (Focus on Parallelization):** `"Aho-Corasick" AND (parallel* OR multithread* OR multi-thread* OR thread* OR concurren* OR hpc OR high-perform* OR high perform*) AND ("multi-core" OR multicore OR "many-core" OR CPU) NOT "distribut systems")`
- **String 2 (Focus on Optimization and Applications):** `"Aho-Corasick" AND (optimi* OR accelerat* OR perform* OR evaluat* OR improve*) AND ("application" OR "implementation" OR "use case"))`

3.2.3 Inclusion (IC) and Exclusion (EC) Criteria

The subsequent criteria were established for the selection of primary studies:
Inclusion Criteria (IC):

- **IC1 (Primary Focus):** The study addresses the Aho-Corasick (AC) algorithm or presents parallelization/optimization techniques demonstrably applicable to AC.
- **IC2 (Parallelism Aspect):** The study investigates, proposes, implements, analyzes, or explicitly discusses the parallelization or concurrent execution of the Aho-Corasick algorithm.
- **IC3 (Target Platform):** The study reports empirical results or analyses about multi-core CPU platforms.
- **IC4 (RQ Relevance):** The study provides information relevant to addressing at least one RQ.
- **IC6 (Language):** Published in English or Portuguese.
- **IC7 (Accessibility):** Full-text availability of the study.
- **IC8 (Publication Window):** Published between January 2015 and May 2025, inclusive.

Exclusion Criteria (EC):

- **EC1 (Irrelevant Focus):** Studies with only superficial mentions of the Aho-Corasick algorithm or pattern matching, lacking a primary focus on its parallelization or optimization.

- **EC2 (Non-CPU Platform Exclusivity):** Studies exclusively focused on non-CPU platforms (e.g., GPU, FPGA) without analysis or insights transferable to multi-core CPU environments.
- **EC4 (Insufficient Detail):** Studies lacking sufficient detail regarding methodology, implementation, or reported results.
- **EC5 (Publication Type):** Opinion pieces, editorials, and non-peer-reviewed gray literature.
- **EC6 (Language Barrier):** Publications in languages other than English or Portuguese.
- **EC7 (Duplication):** Duplicate publications (the most comprehensive or recent version was retained).
- **EC8 (Inaccessibility):** Full-text unobtainable.

3.2.4 Selection Process and Initial Search Outcomes

The study selection was a multi-stage process. Initially, the constructed search strings were executed against the selected digital libraries. The retrieved results were subsequently refined using filters provided by each database (e.g., publication period, document type, language, subject area), as detailed below:

- **ACM Digital Library:** String 1 yielded 122 articles, reduced to 29 following filter application
  String 2 yielded 274 articles, reduced to 66 following filter application
  Filters applied: Publication years: 2015-2025; Document type: Research Article; Publisher: ACM; Title exclusion: GPU; Language: English.
- **Web of Science:** String 1 yielded 30 articles, reduced to 6 after filter application
  String 2 yielded 66 articles, reduced to 20 after filter application.
  Filters applied: Publication years: 2015-2025; WoS Categories: Computer Science (excluding Artificial Intelligence); Title exclusion: GPU; Language: English.
- **Scopus:** String 1 yielded 34 articles, reduced to 6 following filter application
  String 2 yielded 145 articles, reduced to 49 following filter application
  Filters applied: Publication years: 2015-2025 Language: English Subject areas: Computer Science, Engineering; Document types: Article, Conference Paper; Title exclusion: GPU.
- **IEEE Xplore:** String 1 yielded 25 articles, reduced to 2 following filter application
  String 2 yielded 60 articles, reduced to 18 following filter application

Filters applied: Publication years: 2015-2025; Title exclusion: GPU; Language: English.

Following the initial database search and filtering stage, the resultant sets of candidate articles were aggregated.

The filtered outputs from each database and search query were subsequently merged, and duplicate entries were eliminated. Thereafter, a title and abstract screening was performed on the unique articles, and the established inclusion and exclusion criteria were applied. The articles that passed this screening phase (30) proceeded to a full-text review for a conclusive evaluation against the IC/EC. This procedure yielded a selection of Y primary studies designated for comprehensive analysis.

At the end of the selection process, Z studies were considered relevant and included in this systematic review.

3.3 ANALYSIS AND SYNTHESIS FROM SELECTED STUDIES

The 'Z' selected primary studies were critically analyzed in order to extract relevant information to answer the research questions.

3.3.1 Analysis related to RQ1: Techniques and Challenges in Parallelizing Aho-Corasick on Multicore CPUs

The reviewed literature indicates that the Aho-Corasick algorithm, despite its efficiency in sequential implementations for multiple pattern matching, especially in applications such as Network Intrusion Detection Systems (NIDS)...(to be continue)

4

4 METODOLOGIA

A metodologia descreve, de forma clara e detalhada, os procedimentos adotados para o desenvolvimento da pesquisa. É nesta seção que se explica como o estudo foi conduzido, permitindo que outros pesquisadores possam replicar ou compreender os passos realizados.

Devem ser apresentados os seguintes elementos, quando aplicável:

- **Tipo de pesquisa**: definir se é qualitativa, quantitativa ou quali-quantitativa, além de classificar como exploratória, descritiva, explicativa, estudo de caso, pesquisa de campo, bibliográfica, documental, entre outras.
- **Procedimentos metodológicos**: descrever como foi realizado o levantamento de dados, incluindo instrumentos (questionários, entrevistas, observações, análises documentais, softwares, entre outros) e etapas do processo.
- **População e amostra**: identificar o universo da pesquisa, bem como os critérios de seleção da amostra, se houver.
- **Análise dos dados**: indicar quais métodos, técnicas estatísticas ou ferramentas foram utilizados para o tratamento e análise dos dados coletados.
- **Materiais e métodos**: em pesquisas aplicadas ou experimentais, especificar os materiais, equipamentos, ferramentas e procedimentos técnicos utilizados.

A metodologia deve ser apresentada de forma objetiva, precisa e suficientemente detalhada, garantindo a confiabilidade e a possibilidade de reprodução do estudo.

5

5 CONCLUSÃO

A conclusão é a seção final do trabalho, onde o autor retoma os objetivos propostos no início e apresenta as considerações finais com base nos resultados alcançados. Seu papel é responder se o problema de pesquisa foi solucionado e se os objetivos foram atingidos.

Neste capítulo, não se introduzem novos dados, informações ou discussões. O foco está em:

- Retomar brevemente os objetivos do trabalho;
- Apontar se os objetivos foram alcançados;
- Destacar os principais resultados obtidos;
- Apresentar reflexões sobre as limitações do estudo, quando houver;
- Sugerir possíveis desdobramentos, melhorias ou pesquisas futuras sobre o tema.

A conclusão deve ser clara, objetiva e coerente com tudo o que foi desenvolvido ao longo do trabalho, fechando o raciocínio de forma lógica e consistente.

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APPENDICES

Appendix A - Título do apêndice A

Os apêndices são materiais suplementares **elaborados pelo próprio autor**, que contribuem para a compreensão, fundamentação ou detalhamento do trabalho, mas que não são essenciais para o desenvolvimento principal do texto.

Eles podem incluir, por exemplo:

- Questionários e roteiros de entrevistas utilizados na pesquisa;
- Documentos, códigos-fonte, tabelas extensas ou informações complementares;
- Detalhes técnicos ou cálculos que, se inseridos no corpo principal, tornariam a leitura cansativa;
- Materiais que apoiam a metodologia ou os resultados, mas que não são obrigatórios para a compreensão geral.

Cada apêndice deve ser identificado por uma letra maiúscula (Apêndice A, Apêndice B, etc.) e ter um título claro e objetivo. Eles são inseridos após as referências bibliográficas e antes do índice remissivo, se houver.

A inclusão dos apêndices deve ser feita de maneira organizada e referenciada no texto principal, para que o leitor saiba quando e por que consultar esses materiais complementares.

ANNEXES

Annex A - Título do Anexo A

Os anexos são documentos, materiais ou informações que **não foram elaborados pelo autor do trabalho**, mas que servem para complementar, ilustrar ou fundamentar o conteúdo apresentado.

Exemplos comuns de anexos incluem:

- Legislações, normas técnicas ou regulamentos;
- Manuais, folhetos, publicações oficiais;
- Textos, tabelas ou documentos externos que deram suporte à pesquisa;
- Materiais gráficos ou audiovisuais cedidos por terceiros.

Assim como os apêndices, os anexos são organizados em sequência e identificados por letras maiúsculas (Anexo A, Anexo B, etc.), com títulos claros.

Eles são inseridos após os apêndices (quando houver) ou diretamente após as referências bibliográficas, e devem ser referenciados no corpo do trabalho para que o leitor saiba quando consultá-los.

Ensino que
te **conecta**